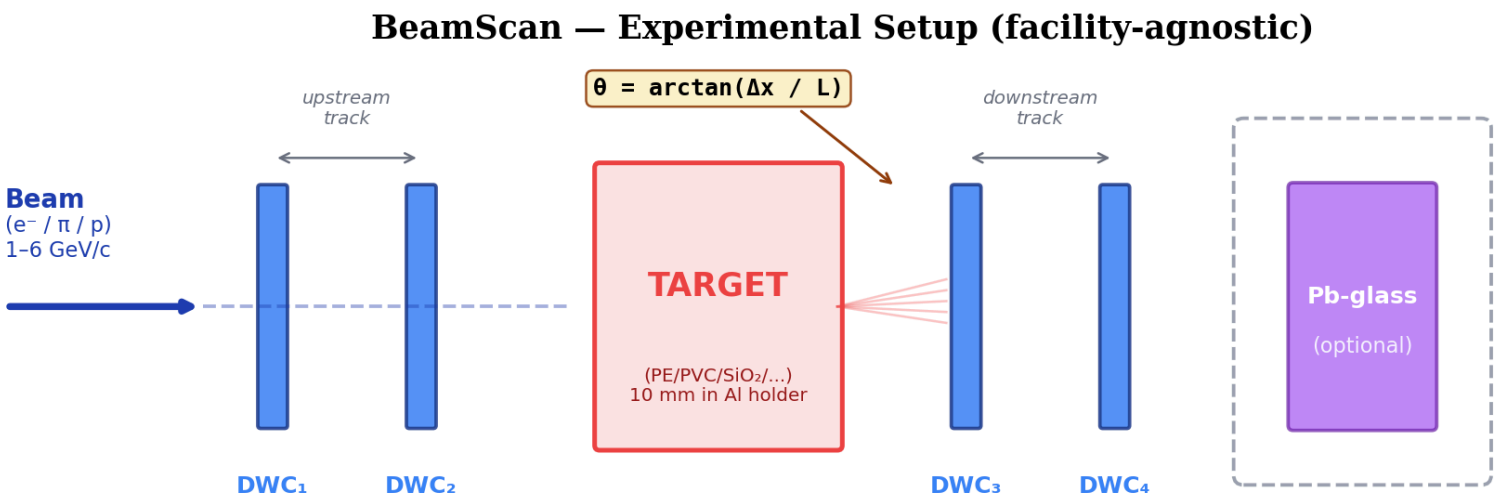
# Technical Deep Dive

What the code actually does — for project owners

7 source files · 2 CI pipelines · 3 analysis scripts · 11 materials

# Geant4 — Detector Geometry




| Component | Material | Size | Purpose |
|---|---|---|---|
| **DWCs** (×4) | Ar/$CO_2$ 80/20 gas (10 mm gap) | 10 × 10 cm² | Measure particle direction before/after target |
| **Target** | Configurable via macro | 10 × 10 cm², variable thickness | Material under study |
| **Calorimeter** | PbO lead glass | 15 × 15 × 30 cm³ | Energy measurement (optional extension) |

# Geant4 — Physics List & Per-Event Data

## Physics List: FTFP_BERT + option4

**FTFP_BERT:** Fritiof + Bertini cascade

- Handles hadronic interactions (p, π, n)
- Nuclear elastic + inelastic scattering
- *This is why G4 gives 12% more than Highland*

**EM option4:** most accurate MCS model

- G4UrbanMscModel for electrons
- G4WentzelVIModel for hadrons
- Step limiter ensures small steps in

## What happens per event

1. `ParticleGun` fires $e^-$ at z = −80 cm
2. Particle traverses DWC1 → DWC2 *(SteppingAction records first hit per plane)*
3. Particle enters **TARGET** → MCS happens
4. Particle exits → DWC3 → DWC4
5. Particle hits Calorimeter → energy deposited
6. `EventAction` computes all angles + momenta

# Geant4 — The 9-Column Ntuple

| Column | Unit | How computed | Used for |
|---|---|---|---|
| `theta3D_mrad` | mrad | `acos(inDir · outDir)` | Full 3D scattering angle |
| `thetaX_mrad` | mrad | `atan2` difference in x-z plane | **Primary classification observable** |
| `thetaY_mrad` | mrad | `atan2` difference in y-z plane | Cross-check / 2D analysis |
| `pIn_GeV` | GeV | Momentum at DWC2 (before target) | Beam energy verification |
| `pOut_GeV` | GeV | Momentum at DWC3 (after target) | Energy loss measurement |
| `deltaP_MeV` | MeV | pIn − pOut | dE/dx spectroscopy |
| `xTarget_mm` | mm | Interpolated x position at target plane | Beam profile |
| `yTarget_mm` | mm | Interpolated y position at target plane | Beam profile |
| `caloEdep_MeV` | MeV | Total energy deposited in lead glass | Particle ID (future) |

**9 columns per event** — enough for MCS classification AND momentum-loss spectroscopy. Students can mine this data for multiple physics analyses beyond the primary classification.

# Geant4 — Multi-Threading & Output Format

## MT-Mode Architecture

Geant4 11.3.2 auto-detects CPU cores. GitHub Actions runner: **4 cores → 4 threads**.

Each thread writes its own CSV file:

```
events_nt_beamscan_t0.csv
events_nt_beamscan_t1.csv
events_nt_beamscan_t2.csv
events_nt_beamscan_t3.csv
```

`ActionInitialization.cc` creates per-thread copies of `RunAction` + `EventAction` + `SteppingAction`.

## WCSV Format (not standard CSV!)

Geant4 writes `#class tools::wcsv::ntuple` headers with metadata, **not a plain CSV**.

Our parser (`analyze_geant4.py`) handles:

- `#separator` line → detects delimiter
- `#column` lines → extracts column names
- Data lines → splits and zips into dicts

**Why this matters:**

`pandas.read_csv()` **fails** on Geant4

# Analysis Pipeline — From Raw Events to Classification

**1. Raw CSV**
θx, θy, θ3D,
pIn, pOut, ΔP,
x, y, Ecalo
per event

→

**2. Core Cut**
Remove tails:
keep central 95%
(2.5→97.5 pctl)
Mitigate nuclear
scattering outliers

→

**3. Gaussian Fit**
σ_core = std(core)
μ_core =
mean(core)
This is the G4
"measured" σ

→

**4. Highland Match**
Load
predictions.csv
Match
material_pGeV
Compute ratio:
G4_σ /
Highland_$\theta_0$

→

**5. Outputs**
Bar plot
Comparison table
CSV data
SUMMARY.md

## Why the Core Cut Matters

MCS produces a **Gaussian core + non-Gaussian tails**. The tails come from: (a) nuclear elastic scattering (large single kicks), (b) nuclear inelastic events (secondaries), (c) delta rays. Highland predicts only the Gaussian core, so we cut the tails for a fair comparison.

# Highland Calculator — The Analytic Engine

## The Formula (PDG 2024)

$$\theta_0 = \frac{13.6\ \mathrm{MeV}}{p\beta c}\sqrt{\frac{x}{X_0}}\left[1 + 0.038 \ln \frac{x}{X_0}\right]$$

### `compute_predictions()` does:

1. Look up $X_0$, $\rho$ from `MATERIALS_DB`
2. Or accept **student-provided** $X_0/\rho$ in YAML *(custom materials!)*
3. Compute $\theta_0$ via Highland formula
4. Estimate $dE/dx \approx 2\ \mathrm{MeV}/(\mathrm{g/cm^2}) \times \rho \times x$
5. Store result: $\theta_0$, dE, $X_0$, $\rho$, $p$, thickness
6. For all material pairs:

## Output Files

**`predictions.csv`** — 11 materials × 2 momenta = 22 rows
Columns: name, g4name, category, $X_0$, ρ, thickness, p, $\theta_0$, dE

**`distributions.png`** — Gaussian curves overlaid for all materials

**`classification.png`** — 2D plot: $\theta_0$(3 GeV) vs $\theta_0$(6 GeV) with 1/p line and cluster annotations

**`SUMMARY.md`** — Table + pair separations + N_events

# CI/CD — Two Pipelines, Two Purposes

## ⚡ Highland CI (~30 sec)

**Triggers:** PR touching `requests/*.yaml`, push to main, or manual dispatch

**Steps:**

1. Checkout repo
2. `pip install pyyaml jsonschema matplotlib`
3. Validate YAML against JSON schema
4. Run `highland_calculator.py`
5. Upload `predictions/` as artifact
6. If PR: **post comment with plots +**

## ⚛️ Geant4 CI (~20 min)

**Triggers:** manual dispatch only *(too expensive for every PR)*

**Steps:**

1. Checkout repo
2. Install Miniforge (conda)
3. `conda install geant4` (11.3.2) + cmake, gcc, matplotlib, scipy
4. `cmake + make beamscan`
5. `generate_macros.py` → 22 `.mac` files
6. `run_all.sh` → runs all 22 simulations

# Macro Generation & Scattering Computation

## `generate_macros.py`

Reads student YAML → writes one `.mac` per material × momentum:

```
# Auto-generated: PE_3.0GeV_10.0mm.mac
/run/initialize
/gun/particle e-
/gun/energy 3.0 GeV
/beamscan/target/material G4_POLYETHYLENE
/beamscan/target/thickness 10.0 mm
/beamscan/output/filename \
  geant4_output/PE_3.0GeV_10.0mm/events
/run/beamOn 2000
```

Plus `run_all.sh`:

```
for f in macros_auto/*.mac; do
```

## The Scattering Angle Computation

`SteppingAction` records first hit per DWC → position + momentum at each plane.

`EventAction::EndOfEventAction` computes:

```
inDir  = (pos[DWC2] − pos[DWC1]).unit()
outDir = (pos[DWC4] − pos[DWC3]).unit()

θ_3D = acos(inDir · outDir)
θ_x  = atan2(out.x, out.z) − atan2(in.x, in.z)
θ_y  = atan2(out.y, out.z) − atan2(in.y, in.z)
```

Only events hitting **all 4 DWCs** are kept (natural acceptance cut).

# Bugs Found & Fixed — Engineering Story

| Severity | Bug | Root Cause | Fix |
|---|---|---|---|
| CRITICAL | **$X_0$ Values Wrong (4 materials)** | PVC, $CaCO_3$, $Al_2O_3$, $Fe_2O_3$ had $X_0$ in g/cm² instead of cm. Highland overpredicted PVC (physically impossible). | Recomputed using Tsai formula + PDG cross-checks |
| HIGH | **PET/PE Substring Collision** | `"PE" in "PET_3.0GeV"` → `True` in Python. Highland matching grabbed PE's value for PET. | `m.startswith(material + "_")` instead of `in` |
| MEDIUM | **Workflow Push Race Condition** | 20-min Geant4 run + concurrent push → rejected. | `git pull --rebase` before `git push` |
| REJECTED | **ChatGPT Review — 8 Regressions** | Reviewed old snapshot. Deleted `ActionInitialization` (breaks MT), removed wcsv parser, reverted to hardcoded filenames, used wrong PE $X_0$. | All changes rejected after analysis |

**Every bug has a commit, a rationale, and a verification.** This is engineering discipline that BL4S reviewers will notice — it shows the team understands what their code does.

# Request Schema — What Students Can Configure

## Full YAML Structure

```yaml
author: "Name"                # required
description: "Research Q"     # required
materials:                    # 1+ entries
  - name: PE                  # display name
    geant4_name: G4_POLYETHYLENE
    thickness_mm: 10          # 1-100 mm
    # Optional: override for custom materials
    X0_cm: 47.9
    rho: 0.94
beam:
  particle: e-        # e-, e+, pi+, pi-
  momenta_GeV: [3.0, 6.0]  # list
num_events: 10000  # per configuration
```

Validated by `schemas/request.schema.json` (Draft 2020-12).

## Named Examples in Repo

**Valentina** — `valentina_pvc_detection.yaml`

PVC vs PE → recycling QC

**Tomás** — `tomas_thickness_study.yaml`

PE at 5, 10, 20 mm → test $\sqrt{(x/X_0)}$ scaling

**Lucía** — `lucia_heritage_study.yaml`

$SiO_2$, $CaCO_3$, $Al_2O_3$, $Fe_2O_3$ → heritage materials

**Sofía** — `sofia_custom_material_example.yaml`

Student provides own $X_0$ and $\rho$ → custom

# End-to-End

Student types → gets publication-ready physics plots back in

📝 YAML → 🔀 PR → 🤖 CI → 📊 Plots → ✅ Merge → ⚛️ Geant4 → 🌐 Website

*¡La física fundamental es para todos!*